

How the digital twin computes one control step

The pipeline that turns a scenario and a policy action into KPIs and a reward — propagation, association, scheduling, reward, and the PPO / Double DQN learning loop — with the formula behind each stage as implemented.

single-agent · per-tier action branching 3GPP TR 38.901 / 38.811 / 38.821
ITU-R P.618 / P.838 / P.839

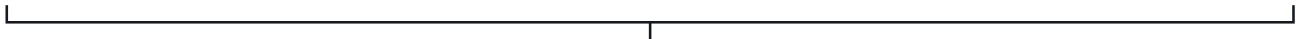
01 Two timescales

Everything runs on a fast **tick** and a slow **control interval**. Each tick ($\Delta t = 0.1 \text{ s}$ by default) advances UE positions along their velocity vectors, recomputes RSRP, and steps the A3/A5 measurement state machine. Once per control interval ($100 \text{ ticks} = 10 \text{ s}$) the policy chooses an action, PRBs are allocated, and the interval's KPIs are aggregated. One `step()` = one control interval; `T` intervals make an episode.

fast loop — mobility, RSRP, A3/A5 timers



$\Delta t = 100 \text{ ms}$



one control interval — 100 ticks = 10 s

policy action → PRB allocation → KPI aggregation

The measurement machinery only exists at tick rate; the policy only acts at interval rate.

02 Offered demand

Each session carries a base rate from its type (eMBB 5, URLLC 1, mMTC 0.02 Mbps). A per-slice load factor $\lambda_s(t)$ scales those rates over the episode. Its shape is one of: `flat` ($\lambda \equiv 1$); `diurnal`, a two-peak day; or a named profile taken from the platform's established trace `synthetic_profiles/all_profiles.csv` (P1 ... P5 — busy hour, eMBB hotspot, URLLC burst, massive IoT, mixed-failure stress), normalised to its own peak. A carrier trace drops into that file without touching anything else.

$$\begin{aligned} \text{diurnal: } \text{shape}(t) &= 0.55 + 0.32 \cdot \sin(\phi - 1.9) + 0.18 \cdot \sin(2\phi - 0.4) & (1) \\ \text{profile: } \text{shape}_s(t) &= \text{trace}_s[t \bmod T] / \max_t \text{trace}_s \\ \phi &= 2\pi \cdot (t \bmod T) / T \end{aligned}$$

$$\begin{aligned} \rho_t &= 0.85 \cdot \rho_{t-1} + \mathcal{N}(0, \sigma^2), \quad \sigma = \text{traffic_noise} & (2) \\ \lambda_s(t) &= \text{shape}_s(t) \cdot (1 + \rho_t) \quad \rightarrow \quad D_i = \text{rate}(\text{type}_i) \cdot \lambda_{s(i)}(t) \end{aligned}$$

mMTC is held less diurnal (a $0.45 + 0.55 \cdot \text{shape}$ floor).

03 Propagation & channel

Per tier, path loss comes from the tier's **morphology** (UMa / UMi / RMa / InH → TR 38.901; NTN → TR 38.811). The terrestrial LOS branch, below the breakpoint distance:

$$PL_{LOS} = 28.0 + 22 \cdot \log_{10}(d_{3D}) + 20 \cdot \log_{10}(f_c) \quad [UMa] \quad (3)$$

LOS and NLOS are blended by the TR 38.901 LOS probability in the linear domain, so a single distance-indexed table per tier is exact:

$$PL = -10 \cdot \log_{10}(p_{LOS} \cdot 10^{-PL_{LOS}/10} + (1-p_{LOS}) \cdot 10^{-PL_{NLOS}/10}) \quad (4)$$

Indoor UEs served by an outdoor cell add the TR 38.901 O2I term (low or high loss). Each link also carries spatially-correlated log-normal **shadow fading** — one value per (session, tier), σ from the TR 38.901 tables ($\approx 6-8$ dB), evolving as a Gudmundson AR(1) process with the distance the UE moves:

$$\begin{aligned} \xi_t &= e^{-\Delta d/d_{corr}} \cdot \xi_{t-1} + \sqrt{1 - e^{-2\Delta d/d_{corr}}} \cdot \sigma \cdot \mathcal{N}(0,1) \quad d_{corr} \approx 50 \text{ m} \\ RSRP &= EIRP_{RE} - PL - L_{O2I} - L_{blockage} - \xi_t \end{aligned} \quad (5)$$

SINR is against thermal noise *and co-channel interference* — the sum of received power from same-band cells other than the server, each weighted by its load ρ_j (activity) and the tier's beam-coupling κ (1.0 omni, 0.35 upper-mid-band, 0.08 mmWave). 12 REs per PRB:

$$\begin{aligned} N_0 &= -174 + 10 \cdot \log_{10}(B_{PRB} \cdot 10^6) + NF \\ I &= 12 \cdot \sum_{j \neq \text{serv, same band}} P_{rx,j}^{RE} \cdot \rho_j \cdot K_j \\ \text{SINR} &= \frac{P_{rx,serv}^{PRB}}{10^{N_0/10} + I} \quad (\kappa = 0 \Rightarrow \text{noise-limited, the old model}) \end{aligned} \quad (5')$$

Non-terrestrial (LEO)

Slant range at elevation ϵ over a spherical Earth ($R = 6371$ km, h = altitude), then free-space loss:

$$d = \sqrt{(R \cdot \sin \epsilon)^2 + 2R \cdot h + h^2} - R \cdot \sin \epsilon \quad (6)$$
$$\text{FSPL} = 92.45 + 20 \cdot \log_{10}(f_{\text{GHz}}) + 20 \cdot \log_{10}(d_{\text{km}})$$

Rain attenuation follows ITU-R P.618 with the P.838 coefficients and P.839 rain height; it is negligible at S-band and dominant at Ka-band:

$$\gamma_R = k \cdot R_{\text{rain}}^\alpha \quad [\text{dB/km}] \quad A_{\text{rain}} = \gamma_R \cdot L_S \cdot r_{0.01} \quad (7)$$

The LEO pass sweeps elevation $\epsilon_{\min} \rightarrow 90^\circ \rightarrow \epsilon_{\min}$; Doppler and its residual after pre-compensation drive inter-carrier interference:

$$f_d = f_c \cdot \dot{r} / c \quad \epsilon_{\text{off}} = \Delta f_{\text{res}} / \text{SCS} \quad \text{ICI/S} \approx \frac{(\pi \cdot \epsilon_{\text{off}})^2}{3} \quad (8)$$

Per-stream spectral efficiency is Shannon, capped, and de-rated for mobility; MIMO multiplies delivered cell capacity:

$$se = \min(\log_2(1 + \text{SINR}), se_{\max}) \cdot \eta_{\text{mob}} \cdot (1 - \text{ICI/S}) \quad (9)$$
$$C_{\text{cell},s} = n_{\text{PRB},s} \cdot B_{\text{PRB}} \cdot se \cdot G_{\text{MIMO}}$$

$$G_{\text{MIMO}} = \max(1, L) \cdot \eta_{\text{MIMO}} \cdot (1.15 \text{ if MU else } 1.0) \quad (10)$$
$$n_{\text{PRB}} = \lfloor 0.95 \cdot \text{BW} \cdot N_{\text{CC}} / (12 \cdot \text{SCS}) \rfloor \quad \text{if not given explicitly}$$

04 Association & handover

Each UE holds a measurement set of its nearest cells (plus the satellite beam, always). A **steering eligibility mask** is applied first: indoor-only cells serve indoor UEs, outdoor-only (the LEO beam) serve outdoor UEs, a tier cannot hold a UE faster than its $v_{\max}$, and the satellite is reserved for $v > v_{\text{NTN}}$ or a best-terrestrial RSRP below the coverage-hole threshold.

A3/A5 events are evaluated on biased levels $M = \text{RSRP} + \text{CIO} + \text{steering bias}$. Entry conditions (TS 38.331), held for the time-to-trigger:

$$\begin{aligned} \text{A3: } & M_n - \text{Hys} > M_s + \text{Off} \\ \text{A5: } & M_s + \text{Hys} < T_1 \quad \wedge \quad M_n - \text{Hys} > T_2 \end{aligned} \tag{11}$$

Off, Hys, TTT, T_1 , T_2 are per session type: URLLC uses a short 40 ms trigger, mMTC a long 1280 ms trigger with a large offset.

A fired handover can still fail. Probability of a late (failed) handover rises with UE speed and falls with the trigger margin:

$$P_{\text{late}} = \text{clip}(0.01 + 0.10 \cdot (v/120) - 0.01 \cdot \Delta_{\text{margin}}, 0, 0.6) \tag{12}$$

Also tracked: admission refusal (target at capacity), speed rejection, ping-pong (return to the previous cell inside a 5 s window), radio-link failure ($M_s < M_{\min}$), and the interruption time — which breaks continuity when it exceeds the class's tolerance.

05 Scheduling & KPIs

Each tier declares which of the three slices it carries. A full terrestrial tier serves `{eMBB, URLLC, mIoT}`; `mmwave` drops `mIoT`; `wifi7_indoor` and `ntn_leo` drop URLLC — a ~25 ms LEO service-link RTT cannot underwrite a 5 ms budget. The policy's per-tier template π_{tier} is masked to that set and renormalised ($\pi_s \leftarrow \pi_s \cdot m_s / \sum \pi \cdot m$), then PRBs are shared equally among the sessions of each slice on the cell. Served rate per session is `min(demand, capacity)`, capped by backhaul. Aggregated per slice:

$$\begin{aligned} \sigma_s &= \text{clip}(\sum \text{served}_s / \max(\sum \text{offered}_s, \varepsilon), 0, 1) \\ \text{SLA}_s &= \mathbb{1}[\text{offered}_s \leq \varepsilon \vee \sigma_s \geq \theta_s] \quad \theta = (0.95, 0.999, 0.99) \end{aligned} \quad (13)$$

Unmet demand is split by cause — the distinction the feasibility check needs:

$$\begin{aligned} \text{gap}_{\text{cov},s} &= \sum \text{demand of sessions with no eligible tier} \\ \text{gap}_{\text{cap},s} &= \sum (\text{demand} - \text{served}) \text{ over attached sessions} \end{aligned} \quad (14)$$

Jain fairness over the three slice satisfactions:

$$F = \frac{(\sum \sigma_s)^2}{N \cdot \sum \sigma_s^2} \quad (15)$$

06 Reward

One scalar per control interval. Satisfaction is weighted toward URLLC; utilisation and fairness are small positive terms; SLA violation, energy, handover churn and lost continuity are penalties.

$$V = \sum_s (1 - \text{SLA}_s) \cdot k_s + 1.5 \cdot L_{\text{risk}} \quad k = (1, 4, 1.5) \quad (16)$$

$$\chi = (n_{\text{ping-pong}} + n_{\text{RLF}}) / N_{\text{sessions}}$$

$$r = w \cdot \sigma + 0.06 \cdot U + 0.04 \cdot F - 0.12 \cdot V - c_E \cdot E_{\text{frac}} - c_{\text{HO}} \cdot \chi - c_C \cdot (1 - C) \quad (17)$$

SYMBOL	MEANING	VALUE
w	slice satisfaction weights (eMBB, URLLC, mMTC)	0.27, 0.48, 0.25
U, F	network utilisation, Jain fairness	—
L_{risk}	URLLC latency risk (near-saturated tiers carrying URLLC)	[0, 1]
c_E	energy price $\cdot E_{\text{frac}}$ (of max)	0.03
c_{HO}	handover-churn price	0.05
c_C	continuity price $\cdot (1 - \text{attached fraction})$	0.10

07 Learning

Observation: a 46-D vector — demand, phase, utilisation, continuity, handover stats, NTN elevation and rain, plus six features per tier. Action is **per tier**: one of nine slice-mix templates for each tier, plus one global steering posture — a multi-categorical action with $n_{\text{tiers}} + 1$ softmax heads on a shared $46 \rightarrow 64 \rightarrow 64$ trunk. Single agent, one update.

PPO

The joint log-probability is the sum over branches; the clipped surrogate acts on the joint ratio, with GAE(λ) advantages.

$$\begin{aligned}\rho_t &= \exp(\sum_k \log \pi_k(a_{k,t}) - \log \pi_{\text{old},t}) \\ \mathcal{L}^{\text{clip}} &= -\mathbb{E}[\min(\rho_t \cdot \hat{A}_t, \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon) \cdot \hat{A}_t)]\end{aligned}\tag{18}$$

$$\begin{aligned}\delta_t &= r_t + \gamma \cdot V(s_{t+1}) \cdot (1-d_t) - V(s_t) \\ \hat{A}_t &= \delta_t + \gamma \cdot \lambda \cdot (1-d_t) \cdot \hat{A}_{t+1} \quad \gamma = 0.95, \lambda = 0.95, \epsilon = 0.2\end{aligned}\tag{19}$$

Double DQN (branching)

Per-tier Q heads; the online net picks each branch's action, the target net scores it, and the per-branch bootstraps are averaged into one shared TD target (BDQN). Huber loss.

$$y_t = r_t + \gamma \cdot (1-d_t) \cdot \frac{1}{K} \sum_k Q_{\text{tgt},k}(s_{t+1}, \arg\max_a Q_{\text{on},k}(s_{t+1}, a))\tag{20}$$

08 Optimisation & feasibility

Coordinate descent over a knob set chosen by objective —

handover · throughput · latency · coverage . Each objective is scored on its own KPI, always net of a session-continuity guard, so nothing can win by stranding sessions:

$$\begin{aligned} \text{guard} &= 2 \cdot (1 - C) + 1 \cdot \text{rate}_{\text{ping-pong}} \\ \text{score}_{\text{lat}} &= -\text{latency} / 5 + 0.5 \cdot \sigma_{\text{URLLC}} - \text{guard} \end{aligned} \quad (21)$$

Knobs: per-class A3/A5 thresholds, per-tier cell individual offset CIO , steering thresholds, per-class $M_{\min}$.

FEASIBILITY VERDICT

Every run reports whether threshold tuning can even help. With $g_{\text{cov}} = \Sigma \text{gap}_{\text{cov}} / \Sigma \text{ offered}$ and $g_{\text{cap}} = \Sigma \text{gap}_{\text{cap}} / \Sigma \text{ offered}$:

```
g_cov > 0.05 → COVERAGE-LIMITED (needs sites / low-band / NTN)
g_cap > 0.10 ∧ headroom < 0.1 · Σ offered → CAPACITY-LIMITED (needs
spectrum / cells)
otherwise → FEASIBLE (tune thresholds)
```

`thresholds_can_help` is true only for FEASIBLE.

09 Peak vs loaded capacity

Two capacity figures are reported. **Peak** assumes full PRB use, every UE at the tier's SE ceiling, no interference — a hardware upper bound. **Loaded** substitutes an interference-limited SE from a hex reuse-1 screening estimate: six first-ring interferers at $\sqrt{3} \cdot r$ from the deployed cell density, each at 50% activity, scaled by κ , averaged over a cell-edge and a mid-cell point.

$$\begin{aligned} r &= \sqrt{A / (N_{\text{cells}} \cdot \pi)} \quad \text{nominal cell radius} \\ \text{SINR}_{\text{loaded}} &= \frac{P_{\text{rx}}(0.7r)}{N_0 + 6 \cdot P_{\text{rx}}(\sqrt{3} \cdot r) \cdot \kappa \cdot 0.5} \quad C_{\text{loaded}} = n_{\text{PRB}} \cdot B_{\text{PRB}} \cdot \text{SE}_{\text{loaded}} \cdot G_{\text{MIMO}} \end{aligned} \quad (22)$$

Coverage is the fraction of area within the link-budget radius of a cell, at the deployed density, with a 1.28σ shadow-fade allowance and a 5% inter-cell edge outage — so a well-covered tier reads $\approx 95\%$, not 100%.

10 Not yet in the model

OPEN

PRACH / random-access overload — no preamble-capacity limit or contention, so an mIoT arrival storm never causes access failure and mIoT / URLLC connections-per-cell are shown but not enforced. **Per-KPI targets as inputs** — eMBB throughput, URLLC latency and mIoT coverage-per-km² are internal constants, not scenario knobs.

Closed since v1: co-channel interference in the SINR (§3), spatially-correlated shadow fading (§3), and trace-driven time-varying offered load (§2).

platform/.